

Practical Assignment: 6

Title: Implement any one of the following Expert System for stock market.

CODE :

```
import tkinter as tk

# ----- Expert System Logic -----
def expert_system(trend, news):
    if trend == "up" and news == "positive":
        return "BUY: Market is strong!"
    elif trend == "down" and news == "negative":
        return "SELL: Market is weak!"
    else:
        return "HOLD: Market uncertain!"

# ----- Main App -----
class App(tk.Tk):
    def __init__(self):
        super().__init__()
        self.title("Stock Expert System")
        self.geometry("500x400")

        self.frames = {}

        for F in (HomePage, ChatbotPage, ExpertPage):
            frame = F(self)
            self.frames[F] = frame
            frame.place(relwidth=1, relheight=1)

        self.show_frame(HomePage)

    def show_frame(self, page):
        frame = self.frames[page]
        frame.tkraise()

# ----- Home Page -----
class HomePage(tk.Frame):
    def __init__(self, master):
        super().__init__(master)

        tk.Label(self, text="Stock Market System", font=("Arial", 18)).pack(pady=20)

        tk.Button(self, text="Chatbot",
                  command=lambda: master.show_frame(ChatbotPage)).pack(pady=10)

        tk.Button(self, text="Expert System",
                  command=lambda: master.show_frame(ExpertPage)).pack(pady=10)

# ----- Chatbot Page -----
class ChatbotPage(tk.Frame):
    def __init__(self, master):
        super().__init__(master)
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self.step = 0
self.trend = ""

tk.Label(self, text="Chatbot", font=("Arial", 16)).pack(pady=10)

self.chat = tk.Text(self, height=12, width=50)
self.chat.pack()

self.entry = tk.Entry(self)
self.entry.pack(pady=5)

tk.Button(self, text="Send", command=self.chatbot_logic).pack()

tk.Button(self, text="← Back",
          command=lambda: master.show_frame(HomePage)).pack(pady=10)

self.chat.insert(tk.END, "Bot: Hello! Click Send to start.\n")

def chatbot_logic(self):
    user = self.entry.get().lower()
    self.chat.insert(tk.END, "You: " + user + "\n")
    self.entry.delete(0, tk.END)

    if self.step == 0:
        self.chat.insert(tk.END, "Bot: Do you want stock advice? (yes/no)\n")
        self.step = 1

    elif self.step == 1:
        if user == "yes":
            self.chat.insert(tk.END, "Bot: Enter trend (up/down)\n")
            self.step = 2
        else:
            self.chat.insert(tk.END, "Bot: Type yes when needed.\n")

    elif self.step == 2:
        self.trend = user
        self.chat.insert(tk.END, "Bot: Enter news (positive/negative)\n")
        self.step = 3

    elif self.step == 3:
        result = expert_system(self.trend, user)
        self.chat.insert(tk.END, "Bot: " + result + "\n")
        self.chat.insert(tk.END, "Bot: Want more advice? (yes/no)\n")
        self.step = 1

# ----- Expert System Page -----
class ExpertPage(tk.Frame):
    def __init__(self, master):
        super().__init__(master)

        tk.Label(self, text="Expert System", font=("Arial", 16)).pack(pady=10)

        tk.Label(self, text="Trend (up/down):").pack()

```

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self.trend_entry = tk.Entry(self)
self.trend_entry.pack()

tk.Label(self, text="News (positive/negative):").pack()
self.news_entry = tk.Entry(self)
self.news_entry.pack()

tk.Button(self, text="Get Advice", command=self.get_result).pack(pady=10)

self.result_label = tk.Label(self, text="")
self.result_label.pack()

tk.Button(self, text="← Back",
          command=lambda: master.show_frame(HomePage)).pack(pady=10)

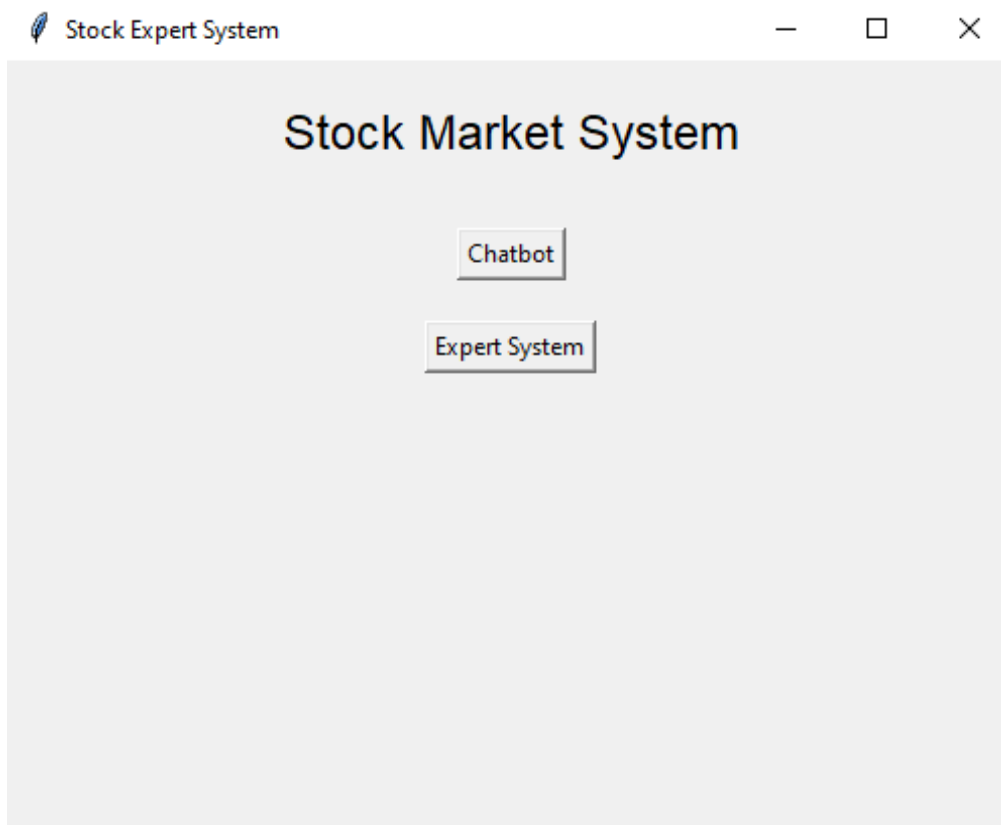
def get_result(self):
    trend = self.trend_entry.get().lower()
    news = self.news_entry.get().lower()

    result = expert_system(trend, news)
    self.result_label.config(text=result)

# ----- Run App -----
app = App()
app.mainloop()

```

OUTPUT :



Chatbot

Bot: Hello! Click Send to start.
 You:
 Bot: Do you want stock advice? (yes/no)
 You: yes
 Bot: Enter trend (up/down)
 You: up
 Bot: Enter news (positive/negative)
 You: positive
 Bot: BUY: Market is strong!
 Bot: Want more advice? (yes/no)
 You: no
 Bot: Type yes when needed.

Send

 Back

Expert System

Trend (up/down):

News (positive/negative):

Get Advice

HOLD: Market uncertain!

 Back